
CERTEN: The Control Layer of Agentic Finance

Technical Whitepaper

A Universal Authorization Protocol for Institutions and Autonomous Agents

Portable Identity. Provable Control. Every Chain.

Control before execution. Proof before movement.

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Forward-Looking Statement. This document describes the design and current testnet implementation of the CERTEN protocol. Capability descriptions reflect either current testnet state or protocol design intent. Specific configuration values (validator counts, batching windows, fee amounts), forward features (Sovereign Exit, mainnet operations, validator economics, and the agentic surface: agent principals, policy-engine signers, and MCP / x402 / agent-identity integrations), and statements about cross-chain coverage beyond currently deployed networks are subject to change as the protocol matures and as third-party security audits inform design refinement. Section 8.4 addresses regulatory interpretation; institutions implementing CERTEN should obtain independent legal counsel. The CERTEN protocol is under active development. Mainnet launch is targeted for Q4 2026, contingent on security audit completion and validator network operational validation.

1. Abstract

Digital assets are controlled by keys. Institutions are governed by identity.

This mismatch creates a structural gap between how assets move on blockchains and how decisions are made within organizations. Existing solutions attempt to bridge this gap by embedding governance into custody platforms or smart contracts, forcing institutions to replicate policy logic in new environments that are difficult to audit, costly to maintain, and fragile under change.

CERTEN introduces a governance layer that separates authority from execution.

Institutional decisions are made by designated authorities (systems, roles, and individuals) within existing environments. CERTEN verifies that the correct authorities have participated, resolves governance deterministically, and produces a cryptographic proof of authorization. This proof can be enforced across any blockchain through execution environments that accept only valid governance outcomes.

Built on Accumulate's identity-based architecture, CERTEN provides a chain-agnostic control layer that preserves institutional governance, enables multi-party coordination, and enforces authorization without requiring assets or logic to be relocated.

CERTEN extends this model with protocol-native proof generation, enabling institutions to demonstrate cryptographic evidence of governance satisfaction, aligning with emerging regulatory requirements such as UCC Article 12 "exclusive control" and MiCA asset segregation.

This same gap now extends to a new class of actor. Autonomous agents are beginning to propose and execute financial actions on-chain, and an agent's transaction can be correctly signed, settled, and final while still falling outside the authority it was permitted to act under. CERTEN is designed to govern agents as scoped principals within the same identity framework it already provides for humans, systems, and roles: an agent proposes an action, and an external policy engine, together with the institution's designated authorities, resolves the authorization. The result is one control layer for both institutional and autonomous finance.

2. Introduction: The Control Problem

Institutional adoption of digital assets has introduced a fundamental mismatch between governance and execution.

Within institutions, control is defined by identity. Decisions are made by systems, roles, and individuals operating under established policies, compliance frameworks, and operational procedures. These processes are distributed, auditable, and deeply integrated into existing infrastructure.

On blockchains, control is defined by possession of private keys.

This mismatch forces institutions into a duplicate policy stack. Governance must be recreated inside custody platforms, smart contracts, or application layers, separate from the systems where decisions

are actually made. As a result, institutional control becomes fragmented across environments that cannot be easily reconciled.

Enterprise computing solved this class of problem decades ago with LDAP, PKI, and OAuth: standard ways to express who someone is, what they are permitted to do, and how to delegate authority without surrendering credentials. Digital assets have no cross-chain equivalent. There is no LDAP for a treasury, no PKI for a permission, no OAuth for an agent.

The Current Options and Their Limitations

Custodial platforms centralize governance in vendor infrastructure. While operationally convenient, they introduce vendor dependency, limit independent verification, and rely on contractual assurances rather than cryptographic proof.

Smart contract wallets embed governance into execution code. While transparent, they require governance to be rewritten as static logic, making systems brittle, expensive to evolve, and vulnerable to coordination-layer attacks.

Neither approach preserves institutional governance as it exists.

Why This Matters Now

UCC Article 12, now being adopted across U.S. states, defines exclusive control of a controllable electronic record, a standard that contemplates technical means of enforcement. MiCA introduces parallel expectations for asset segregation and supervisory auditability.

Existing approaches cannot produce independently verifiable proof that governance requirements were satisfied prior to execution.

CERTEN establishes control prior to execution and produces verifiable proof of that control.

The Agentic Dimension

This mismatch is now being magnified by autonomous software. As institutions begin delegating financial operations to AI agents, authority becomes distributed across humans, systems, and agents that can act continuously and without direct supervision. An agent can hold a key and produce a valid signature, but possession of a key has never been the same as the authority to act. The control problem does not change when the actor is a model rather than a person; it becomes more acute, because the actor is faster, more autonomous, and harder to constrain after the fact. Authorization must be established before the action and remain provable after it. This is the same property CERTEN provides for institutional governance, extended in design to a new principal type.

3. Design Principles

CERTEN is designed as a governance protocol that aligns digital asset control with institutional decision-making, preserving how authority is defined, exercised, and audited within real-world financial systems while enabling deterministic enforcement across decentralized execution environments.

At its foundation, CERTEN enforces a strict separation between authority resolution and execution. In existing blockchain systems, execution environments often determine whether an action is valid by evaluating embedded logic or verifying signatures tied directly to asset control. This model conflates decision-making with execution and forces governance to be reimplemented inside systems that were not designed to express institutional policy.

CERTEN inverts this model. Authority must be fully resolved prior to execution, and execution environments must not participate in determining whether an action is valid. Instead, execution is conditioned solely on the presence of a valid governance proof. This ensures that governance is completed before any asset movement is possible and that enforcement remains consistent regardless of where assets reside.

Authority within CERTEN is defined by identity rather than by possession of cryptographic keys alone. Institutions operate through structured roles, systems, and delegated responsibilities, and these structures form the basis of governance. CERTEN reflects this reality by mapping authority to identities representing roles, systems, and organizational functions, enabling governance to be expressed in terms aligned with institutional structure rather than simplified cryptographic ownership.

Institutional policy remains within existing systems of record. Compliance evaluations, risk assessments, credit approvals, and operational workflows are not migrated into CERTEN or rewritten in smart contracts. Instead, CERTEN verifies that these processes have occurred by requiring participation from designated authorities. The system does not interpret or evaluate policy logic; it verifies that the correct authorities have produced cryptographic attestations of participation. In this model, participation becomes the proof that policy has been satisfied.

All governance outcomes must be deterministic and independently verifiable. Every authorization produces a cryptographic result that can be validated without reliance on intermediaries, vendor systems, or internal records. This ensures that governance is not only enforced, but also provable to auditors, counterparties, and regulators.

Execution environments are intentionally minimized. By removing governance logic from execution, CERTEN reduces attack surface, eliminates the need for complex contract logic, and enables consistent enforcement across heterogeneous blockchain environments. Execution components become simple verification gateways that accept or reject transactions based solely on proof validity.

The system is designed to preserve institutional independence and adaptability. Governance structures can evolve without requiring contract upgrades, asset migration, or dependency on a specific vendor or platform. Authorities can be reassigned, thresholds adjusted, and policies updated

through identity-layer configuration rather than code modification, allowing institutions to respond to regulatory changes and operational requirements without disrupting asset control.

CERTEN also incorporates jurisdictional and regulatory requirements directly into governance. Authority can be defined in terms of jurisdiction, enabling regulatory participation to be enforced as a condition of authorization. This allows institutions to embed compliance requirements within governance structures rather than relying on external processes that cannot be enforced at execution.

Finally, CERTEN is designed to produce cryptographic proof of governance as a first-class outcome. Governance is not an internal process or a system log; it is a verifiable, portable artifact that can be enforced across any execution environment. This enables institutions to demonstrate control, satisfy regulatory requirements such as UCC Article 12, and coordinate multi-party workflows across chains without duplicating governance logic.

Together, these principles ensure that governance is resolved where decisions are made, verified through identity-based participation, and enforced universally through cryptographic proof.

4. System Overview

CERTEN separates institutional governance into three distinct layers: the Authority Layer, CERTEN Core, and Policy Vaults. This separation creates a clean architectural boundary between decision-making, governance resolution, and execution enforcement, allowing each function to operate independently while remaining cryptographically linked.

This model can be understood as a structural division between a governance system that resolves authority and an execution system that enforces outcomes. Governance is performed once and enforced everywhere, eliminating the need to replicate policy logic across multiple systems, contracts, or custodial platforms.

4.1 Authority Layer

The Authority Layer consists of the external systems, roles, and individuals responsible for institutional decision-making.

These authorities include compliance systems performing AML and KYC validation, risk engines evaluating exposure and collateralization, treasury systems managing asset movement, operational workflows coordinating approvals, and human participants acting within defined roles. In addition, external authorities such as third-party compliance providers, data oracles, and specialized regulatory services may participate as part of the governance process.

Each authority operates within its existing environment and evaluates policy according to its own logic. CERTEN does not replace, replicate, or reinterpret institutional policy. Decisions such as whether a transaction meets compliance requirements, satisfies risk thresholds, or adheres to internal controls are performed entirely within the systems that already govern those processes.

Authorities participate in governance by producing cryptographic signatures tied to their identities. These signatures do not represent transaction execution, but rather attest to the completion of decision-making within the authority's domain. In this way, CERTEN captures the outcome of institutional processes without requiring those processes to be reimplemented.

CERTEN integrates these authorities by mapping them into its identity framework. Each authority is associated with a corresponding identity, allowing its participation to be verified as part of a governance process. This mapping enables institutions to reflect their organizational structure directly within the governance system, preserving the relationship between roles, responsibilities, and authority.

Unlike custodial platforms that require institutions to migrate governance into proprietary systems, CERTEN allows institutions to retain their existing infrastructure. This eliminates the need to duplicate policy logic, reduces integration complexity, and avoids introducing new points of failure into critical decision-making processes.

4.2 CERTEN Core

CERTEN Core resolves governance into a deterministic and verifiable decision.

It aggregates cryptographic participation from authorities mapped within the identity framework and evaluates whether all required identities have participated and whether defined thresholds have been satisfied. This evaluation is performed using Accumulate's identity-based architecture, where identities, signature hierarchies, and authorization structures are explicitly defined.

CERTEN Core does not evaluate institutional policy logic. It does not determine whether a compliance check is correct or whether a risk assessment is valid. Instead, it verifies that the authorities responsible for those decisions have participated according to the defined governance structure. This preserves a clear separation between decision-making and verification.

Governance is expressed through structured identity hierarchies using Key Books and Key Pages, which define required categories of approval and valid methods of satisfying those approvals. These structures enable the expression of complex governance requirements, including role-based approvals, quorum thresholds, hybrid human and system validation, and conditional participation paths.

Once all required participation has been collected, CERTEN Core resolves governance deterministically. The result is a governance decision that is independent of execution environment, asset location, or application interface.

This decision is represented as a cryptographic proof that encodes the fact that all required authorities have participated and that all governance requirements have been satisfied. The proof serves as a portable authorization artifact that can be consumed by execution environments across any blockchain.

4.3 Policy Vaults

Policy Vaults act as enforcement gateways on destination chains, ensuring that governance outcomes are applied consistently at the point of execution.

A Policy Vault receives a transaction request accompanied by a governance proof generated by CERTEN Core. The vault verifies the proof and executes the transaction only if the proof is valid. Any transaction that does not include a valid proof, or includes an invalid or incomplete proof, is rejected.

Policy Vaults are required for enforcement. They ensure that no transaction can bypass the governance process, regardless of how the transaction is initiated. Whether a transaction originates from a wallet, application, or automated system, it must pass through a Policy Vault to be executed.

Importantly, Policy Vaults do not evaluate policy logic or interpret governance rules. They do not contain embedded approval conditions, compliance checks, or decision-making processes. Their sole function is to verify proofs and enforce execution outcomes. This keeps execution environments simple, auditable, and resistant to complexity-driven vulnerabilities.

Existing wallets and applications integrate with CERTEN by routing transactions through Policy Vaults. In this model, the wallet manages user interaction and key custody, while CERTEN governs authorization. A transaction is initiated through the wallet, submitted for governance resolution, and then executed through the Policy Vault once a valid proof has been obtained.

This creates a clear operational flow: wallets initiate actions, authorities evaluate and approve, CERTEN resolves governance, and Policy Vaults enforce execution.

By separating these functions, CERTEN enables governance to be performed once and enforced universally across all supported chains. This eliminates duplication of governance logic, reduces operational complexity, and ensures consistent enforcement regardless of execution environment.

5. Architecture

CERTEN implements its governance model using an identity-based architecture built on Accumulate. The system maps external authorities into a deterministic identity framework, resolves authorization through structured signature participation, and produces verifiable cryptographic proofs that can be enforced across heterogeneous execution environments.

This architecture separates governance from execution at a protocol level. Governance is performed within an identity-layer system designed to express institutional authority, while execution occurs within minimal verification environments on destination chains. The linkage between these layers is established through cryptographic proof, which serves as a portable and deterministic representation of authorization.

5.1 Identity Framework (ADIs and Authority Mapping)

Accumulate Digital Identifiers (ADIs) are blockchain-anchored identities that serve as the persistent root of institutional authority, enabling governance decisions to be expressed, verified, and enforced independently of any single system, application, or custody provider.

CERTEN leverages these identities to represent institutional entities, governance domains, systems, roles, and individuals within a unified identity framework.

An ADI serves as the cryptographic root of an institution's governance structure. It defines the hierarchy through which authority is expressed and enforced, including the relationships between roles, required approvals, and participation thresholds. This structure mirrors institutional organization rather than abstract cryptographic ownership, allowing governance to be expressed in terms that align with real-world operational models.

Each authority in the Authority Layer is mapped to an identity within the CERTEN framework. These identities are organized into hierarchical structures that define authorization requirements for specific actions. External systems such as compliance engines, risk services, and operational workflows are integrated as first-class participants through their associated identities, enabling them to contribute directly to governance without modification of their internal logic.

Key Books and Key Pages provide the structural primitives through which governance is expressed. These constructs allow institutions to define both the categories of required approval and the acceptable methods of satisfying those approvals, enabling complex governance structures to be represented deterministically without embedding logic into execution environments.

The same framework accommodates autonomous agents without modification. An AI agent is represented as a principal within an ADI hierarchy, identical in kind to a human role or an external system. The agent occupies a defined position in a Key Page, with its own thresholds, limits, and expiry, and is bound by the other authorities (including policy engines) that participate in the same

structure. In this model an agent is a scoped identity, not a privileged one. Section 5.5 details the agentic configuration.

5.1.1 Role-Based Authority Mapping

CERTEN's identity framework is designed to reflect both organizational structure and functional authority.

Authorities are defined as roles rather than as static cryptographic keys. Roles represent functional responsibilities such as compliance approval, risk validation, treasury authorization, or executive oversight. Individuals and systems are assigned to these roles according to institutional policy, and their authority is derived from that assignment rather than from key ownership alone.

Key Books map directly to roles, defining which categories of authority must participate in governance. Key Pages define how participation within a role can be satisfied, including thresholds, alternative participants, and fallback mechanisms.

For example, a compliance role may include both an automated AML system and a designated compliance officer. A risk role may include a quantitative risk engine and a senior risk manager. A treasury role may include a set of executives operating under a quorum requirement.

A transaction requiring compliance and risk approval does not depend on specific individuals or systems. It depends on participation from the required roles. Authority is therefore dynamic and context-dependent, allowing institutions to maintain continuity of governance even as personnel, systems, or organizational structures change.

5.1.2 Authorization Structure: Key Books and Key Pages

CERTEN expresses governance requirements through the composition of Key Books and Key Pages, enabling deterministic evaluation of complex authorization structures.

Key Books define the categories of approval required for a given action. Each Key Book represents a distinct role or functional domain, such as compliance, risk, treasury, or executive authorization. All Key Books associated with an action must be satisfied for governance to be complete.

Key Pages define the acceptable methods of satisfying each category of approval. Each Key Page specifies one or more valid participation patterns, including threshold requirements, alternative signers, and fallback paths. A Key Book may contain multiple Key Pages, any of which can satisfy the requirement for that category.

This structure results in a conjunctive evaluation across Key Books and a disjunctive evaluation within Key Pages. All required categories must be satisfied, but each category may be satisfied through one of several valid participation patterns.

For example, a transaction may require compliance approval, risk approval, and treasury authorization. Compliance approval may be satisfied by either an automated system or a human officer. Risk approval may require both a system-generated validation and human confirmation. Treasury authorization may require a quorum of senior executives.

This approach enables institutions to express governance requirements that include role-based approvals, quorum thresholds, hybrid human and system validation, conditional participation, and escalation paths. These requirements are expressed declaratively within the identity framework and evaluated deterministically without embedding logic into execution environments.

5.1.3 Dynamic Authority Assignment

Because authority is mapped to identities through ADIs, institutions can modify governance structures without requiring changes to execution environments.

Participants can be added or removed from roles, systems can be replaced, and organizational changes can be reflected immediately within the identity framework. These updates do not require smart contract modification, redeployment, or asset migration.

This capability allows governance to evolve independently of execution, preserving continuity of control while enabling adaptation to regulatory changes, operational requirements, and organizational restructuring.

5.1.4 Policy Vault Alignment

Policy Vaults are mapped to governance requirements through the identity framework. Each vault is associated with a set of required roles and authorization structures defined at the ADI level.

Execution is permitted only when all required roles have been satisfied, sufficient participation thresholds have been met, and governance has been resolved by CERTEN Core. This ensures that enforcement reflects institutional intent as defined within the identity framework rather than static configuration embedded within execution environments.

5.2 Signature Participation Model

CERTEN resolves governance through structured signature participation rather than through conditional execution logic.

In traditional systems, execution environments evaluate conditions to determine whether an action is valid. In CERTEN, validation is performed through participation. Authorities signal completion of their decision-making processes by producing cryptographic signatures, and governance is resolved by verifying that the required set of signatures has been collected.

5.2.1 Intent as the Unit of Authorization

Governance in CERTEN operates on a defined unit of work referred to as an Intent.

An Intent represents a proposed action requiring authorization. It includes a complete description of the action to be performed, the target assets and execution environments, and the governance requirements that must be satisfied. Intents are written directly to the identity layer, where they become immutable records associated with the governing ADI.

Authorities do not sign transactions directly. They sign the Intent. This ensures that all participants are authorizing the same action and that governance is resolved prior to execution.

Because Intents are recorded within the identity layer, they eliminate the need for off-chain coordination mechanisms such as file exchange, messaging systems, or intermediary services. All participants interact with the same canonical representation of the proposed action, reducing the risk of modification, miscommunication, or interception.

5.2.2 Participation as Proof

Governance is resolved through participation rather than through evaluation of policy logic.

Each Intent defines a required set of participating authorities, expressed through Key Books and Key Pages. Authorities participate by signing the Intent after completing their respective evaluations within their own systems.

CERTEN Core aggregates these signatures and verifies that all required identities have participated and that all thresholds have been satisfied. The system does not evaluate the correctness of the underlying decisions. Participation itself serves as the proof that policy requirements have been met.

This model ensures that governance reflects real institutional processes while remaining verifiable at the protocol level.

5.3 Governance Resolution

Once the required signatures have been collected, CERTEN Core resolves governance deterministically.

The system verifies the authenticity of each participating identity, the validity of each signature, the satisfaction of threshold requirements within each Key Page, and the completion of all required Key Books. This evaluation is performed using the identity structures defined within the ADI, ensuring that governance resolution is consistent, repeatable, and independent of execution environment.

If all requirements are satisfied, CERTEN Core produces a governance decision. This decision represents the completion of institutional authorization and is independent of any specific blockchain, application, or custody system.

The governance decision is then transformed into a cryptographic proof that encodes the outcome of the authorization process. This proof serves as a portable artifact that can be verified by execution environments across any supported network.

5.4 Proof Generation

Following governance resolution, CERTEN generates a compact cryptographic proof representing the authorization outcome.

The proof includes commitments to identity participation, verification of threshold satisfaction, and a reference to the governance state within the identity layer. It is designed to be portable, efficient to verify, and independent of the complexity of the underlying governance structure.

5.4.1 Proof Pipeline

CERTEN implements a multi-phase proof pipeline that establishes an unbroken chain of trust from the original Intent through governance resolution to execution on the destination chain.

The pipeline verifies four fundamental properties:

First, that the Intent exists and is included within a confirmed state of the identity-layer blockchain. This ensures that the action being authorized is canonical and immutable.

Second, that the state containing the Intent and associated signatures is valid, having been produced by a Byzantine Fault Tolerant validator set with a verifiable chain of trust to genesis. This ensures that governance resolution is based on a valid and consensus-backed state.

Third, that the correct governance authority authorized the Intent. This includes verification that all required Key Books were satisfied, that thresholds within Key Pages were met, and that all participating identities are valid according to the defined governance structure.

Fourth, that a supermajority of validators attested to the correctness of the proof. This attestation is performed using BLS12-381 aggregate signatures, requiring participation from at least two-thirds of the validator set.

5.4.2 Cryptographic Construction

Proofs are constructed using a combination of signature aggregation and zero-knowledge verification.

Validator attestations are aggregated using BLS12-381 signatures, allowing multiple validator signatures to be combined into a single compact representation. This reduces the size of the proof and enables efficient verification on destination chains.

The aggregated signature and associated validation data are then verified using a Groth16 zero-knowledge proof system. This allows the destination chain to verify the correctness of governance resolution and validator attestation without re-executing the underlying computation.

The resulting proof is constant in size and complexity regardless of the number of participating authorities, the depth of the governance hierarchy, or the number of validators involved.

5.4.3 Verification Characteristics

Proof verification on destination chains has a predictable and bounded cost.

On EVM-compatible chains, verification is designed to require a fixed amount of gas regardless of governance complexity or validator count. This is intended to provide a consistent cost model that is independent of the number of signers or the structure of the authorization hierarchy.

This contrasts with traditional multi-signature approaches, where cost scales with the number of participants and the complexity of the governance logic embedded within execution contracts.

5.4.4 Batching and Anchoring (Introduced)

To optimize cost and throughput, proofs are anchored to destination chains using batching mechanisms.

Multiple governance events can be aggregated into a single commitment through the construction of a Merkle tree. The root of this tree is submitted to the destination chain, and individual proofs include inclusion paths linking them to the committed root.

This allows many governance events to share a single anchoring transaction, reducing cost while preserving verifiability.

Time-sensitive transactions may bypass batching through immediate anchoring, allowing proofs to be submitted independently with lower latency.

Full anchoring mechanics and operational modes are detailed in the subsequent sections.

5.5 Agentic Finance: Agents as Scoped Principals

Autonomous agents are becoming financial actors: proposing trades, moving treasury, and executing payments without direct human initiation. The agentic capabilities described in this section are a

design application of CERTEN's existing identity primitives. They require no change to the underlying protocol, and the agent-facing surface that exposes them is part of the development roadmap, not yet implemented. In this design, CERTEN governs an agent as a new principal type within the identity framework already described. An agent is modeled as a scoped principal occupying a position in an ADI's authorization structure, not a privileged one.

The agent proposes; the authority layer disposes. An agent controls its own segregated on-chain account, but it holds only one position in that account's authorization structure and cannot move assets on its own. The agent proposes an Intent (Section 5.2.1); it cannot resolve governance on its own. A policy engine participates as a signer in the same Key Page: when the agent proposes an action, the policy engine evaluates it against the account's policy and either signs or withholds. Authorization is the conjunction of the agent's proposal and the participation of the policy engine and any human authorities. The agent's reasoning is non-deterministic; the authority that bounds it is deterministic, on-chain, and scoped. This is the signature-participation model of Section 5.2, with the agent and the policy engine as two participating authorities.

The policy engine is a signer, not part of CERTEN. The policy engine that bounds an agent may be the institution's own policy system, a third-party compliance or risk service, or an external AI specialized in fraud, sanctions, or exposure evaluation. CERTEN provides the segregated accounts and the signer hierarchy; the policy logic lives in the signer. This preserves the protocol's core separation (Section 4.1): CERTEN verifies that the required authorities participated and does not evaluate the policy itself.

Hierarchical containment. Because authority is expressed through ADIs and Key Books and Pages, an agent's authority can be scoped, delegated, and revoked structurally:

- An agent can be granted a bounded sub-account, a constrained sub-ADI, for a specific task, with its own limits and expiry. This provides hierarchical containment of a sub-agent.
- An agent can delegate a narrower scope to another agent for a specific action.
- Authority can be revoked from above at any time by the governing ADI, taking effect immediately and without redeploying execution code (Section 5.1.3).

Accountability. Every agent action is an Intent recorded in the identity layer and proven by participation. Accountability resolves to the human or organization that delegated the authority, scoped cryptographically by the ADI tree. This gives agentic finance a determinate answer to the liability question: not "who ran the model," but "under whose authority, and within what scope, was this action permitted." The answer is provable rather than contractual.

Standards interoperability (roadmap). The standards landscape is converging on the same model. In February 2026, NIST launched its AI Agent Standards Initiative to define how autonomous agents authenticate, are authorized, and interoperate across enterprise systems, including adapting existing

identity and authorization frameworks to agents. The model described in this section, in which each agent is an independent identity-bearing principal with scoped, revocable authority and an auditable action history, is aligned with the direction of that work. The protocol is designed to interoperate with the emerging agent standards as the agent-facing surface matures. These interfaces are part of the development roadmap and are not yet implemented. A Model Context Protocol (MCP) interface is designed to expose authorization, scope-checking, delegation, and revocation to agent runtimes as native tools. The protocol is designed to authorize machine-native payment flows such as x402 by validating the authority chain before settlement. Verifiable agent-identity standards such as ERC-8004 are designed to be consumable as inputs to authorization policy. Today the underlying primitives exist (ADIs, segregated accounts, signature participation, and the authorization proof described in Section 5.4); the agent-facing integrations that expose them to agent runtimes are forthcoming.

6. Protocol Flow (Transaction Lifecycle)

The CERTEN protocol defines a deterministic lifecycle through which institutional governance is translated into enforceable execution across blockchain environments.

The process begins with the creation of an Intent. An application or wallet interface constructs a representation of the proposed action, including the target assets, execution environment, and governance requirements. This Intent is written directly to the identity layer under the governing ADI, where it becomes an immutable and canonical record of the proposed transaction. All authorities sign the same canonical Intent, ensuring that governance is resolved over a single, shared representation of the action.

Once the Intent is recorded, designated authorities evaluate the proposed action within their respective systems. Compliance systems perform regulatory checks, risk engines evaluate exposure, and human participants review and approve according to institutional procedures. Each authority, upon completing its evaluation, produces a cryptographic signature tied to its identity, attesting to its participation.

These signatures are collected and aggregated by the validator network. CERTEN Core evaluates the aggregated participation against the governance structure defined by Key Books (required authorities) and Key Pages (valid satisfaction paths), verifying that all required authorities have participated and that all thresholds have been satisfied.

When governance requirements are met, CERTEN Core resolves the Intent and generates a cryptographic proof representing the authorization outcome. This proof encodes the fact that the correct authorities have approved the action and that governance has been completed according to institutional policy.

The transaction, along with the governance proof, is then submitted to the execution layer on the destination chain. Execution can only occur through a Policy Vault, ensuring that no transaction can bypass governance enforcement. The Policy Vault verifies the proof using on-chain verification logic, including validation of the aggregated validator signature and any associated zero-knowledge proof components.

If the proof is valid, the execution contract performs the requested action, such as transferring assets, updating positions, or interacting with other contracts. If the proof is invalid or missing, the transaction is rejected, and no state change occurs.

This lifecycle ensures that authority is resolved prior to execution and that execution is contingent upon verifiable governance. Every step of the process, from Intent creation to final execution, is recorded, verifiable, and resistant to tampering.

The result is a system in which governance is performed once, verified deterministically, and enforced consistently across all execution environments.

7. Security Model

CERTEN's security model is based on separation of concerns, cryptographic verification, deterministic governance resolution, and explicitly defined trust boundaries. Rather than concentrating control within a single system, CERTEN distributes responsibility across distinct layers, each with clearly defined roles and constraints.

At the Authority Layer, decision-making remains within institutional systems and processes. Compliance engines, risk systems, and human approvers are responsible for evaluating whether an action should be authorized. CERTEN does not attempt to validate the correctness of these decisions. Instead, it verifies that decisions were made by the correct authorities according to the defined governance structure. This ensures that institutional policy remains the source of truth while enabling cryptographic verification of participation.

The identity layer enforces governance structure through ADIs, Key Books, and Key Pages. These structures define who must participate and how participation is satisfied. Because governance is expressed declaratively within this framework, authorization requirements are deterministic and cannot be bypassed without modifying the identity-layer configuration. This eliminates ambiguity in how authority is evaluated and ensures that governance outcomes are consistent across all execution environments.

The validator network provides trust-minimized computation and consensus. Operating under a Byzantine Fault Tolerant model, the network ensures that governance resolution and proof generation are correct even in the presence of adversarial behavior. No single validator can produce a valid proof

independently. Proof generation requires participation from a supermajority of validators, and incorrect behavior can be detected and addressed through governance mechanisms within the network.

Cryptographic proofs serve as the enforcement boundary between governance and execution. A valid proof attests that all required authorities have participated and that governance requirements have been satisfied. Execution environments rely exclusively on proof verification and do not evaluate governance logic directly. This prevents execution-layer compromise from bypassing governance, as any transaction lacking a valid proof is rejected.

Execution environments are intentionally minimal and do not contain embedded governance logic. This significantly reduces attack surface compared to traditional smart contract systems, where complex logic introduces potential vulnerabilities. By limiting execution contracts to proof verification and transaction execution, CERTEN ensures that enforcement behavior is predictable, auditable, and resistant to manipulation.

CERTEN eliminates reliance on off-chain coordination mechanisms. Intents are written directly to the identity layer, and all participation occurs through cryptographic signatures associated with those intents. This removes intermediate steps where transactions might otherwise be modified, intercepted, or misrepresented. The canonical representation of an action is immutable and shared across all participants, ensuring consistency throughout the governance process.

The system is resilient to compromised interfaces and wallets. Because authorization depends on satisfying governance requirements rather than on individual keys, a compromised interface cannot unilaterally authorize a transaction. Even if a subset of keys is exposed, execution remains impossible unless all required authorities participate according to the defined governance structure.

CERTEN also provides a path toward protocol-independent recovery through the planned Sovereign Exit mechanism. This mechanism introduces an alternative high-threshold signing path that allows institutions to regain direct control over assets in the event of network failure or systemic disruption. A time-locked safeguard is designed to ensure that this path cannot be abused under normal operating conditions, while still enabling recovery if the governance layer becomes unavailable.

Finally, the separation of governance from execution ensures that institutions retain control over their governance structures independent of any specific implementation or service provider. Governance can be reconfigured, migrated, or recovered without dependency on execution environments or third-party systems.

Taken together, these properties create a security model in which authority is explicitly defined, participation is cryptographically verifiable, governance is deterministically resolved, and execution is strictly constrained by proof-based enforcement.

8. Regulatory Alignment

CERTEN is designed to align digital asset control with emerging regulatory frameworks by making authority explicit, verifiable, and enforceable at the protocol level. Rather than relying on contractual assurances, internal records, or vendor-generated reports, CERTEN is designed to produce cryptographic evidence of governance that can be independently verified by regulators, auditors, and counterparties.

8.1 UCC Article 12: Exclusive Control

UCC Article 12 establishes a new legal standard for control of digital assets, defining exclusive control of a controllable electronic record. The standard is generally understood to contemplate technical means of control, characterized by three core elements: the ability to enjoy the benefits of the asset, the ability to prevent others from exercising control, and the ability to transfer the asset.

CERTEN's architecture is designed to map to these requirements.

Exclusive control is designed to be established through the identity-layer governance structure. The ADI defines the only valid authorization hierarchy capable of producing a governance proof. No other entity, including CERTEN itself, can authorize transactions without satisfying the defined governance requirements.

Prevention of unauthorized access is enforced at the execution layer. Policy Vaults will not execute any transaction unless accompanied by a valid governance proof. Because proof generation requires participation from the defined authorities, no unauthorized party can move assets without satisfying governance.

Transfer capability is preserved through the ability of the governing ADI to initiate and authorize transactions. Institutions retain full operational control over assets through their governance structures.

CERTEN is also designed to address the "system" requirement under UCC Section 12-105 by providing a protocol-native mechanism for demonstrating control. Governance decisions are recorded, resolved, and proven within a deterministic system, allowing institutions to produce cryptographic evidence of control rather than relying on contractual claims.

In collateralized finance scenarios, CERTEN enables automated enforcement of control through mechanisms such as programmatic revocation of authority. For example, a lender's governance structure can remove a borrower's signing rights in response to a trigger event, achieving real-time enforcement of control conditions without manual intervention.

8.2 MiCA Article 75: Asset Segregation and Safekeeping

Under MiCA Article 75, crypto-asset service providers that provide custody are generally required to maintain segregated asset safekeeping, provide sufficient records for supervisory enforcement, and ensure that clients retain access to their assets without undue barriers.

CERTEN is designed to address these requirements through its architecture.

Segregation is designed to be achieved through the use of distinct execution-layer accounts governed by separate identity-layer authorities. Each institutional client can be associated with its own governance structure and execution account, ensuring that assets are not commingled within omnibus custody models.

Auditability is provided through the identity layer, where all governance events are recorded immutably. Every authorization decision, participation event, and proof generation step is preserved as part of the system state, enabling complete reconstruction of governance history.

Client access is intended to be preserved through the separation of governance from execution and the planned Sovereign Exit mechanism. Institutions retain the ability to access and control assets independently of service providers, ensuring that access is not restricted by operational dependencies.

8.3 Jurisdictional Control and Data Residency

CERTEN enables jurisdictional requirements to be expressed directly within governance structures, allowing regulatory compliance to be enforced as part of authorization rather than as an external or advisory process.

Jurisdiction is treated as an explicit authority within the identity framework. Governance structures can require participation from authorities associated with specific jurisdictions, ensuring that regulatory approval is a prerequisite for execution. For example, a transaction involving assets subject to regional regulation may require authorization from a jurisdiction-specific compliance authority before it can proceed.

Institutions can maintain localized control by assigning jurisdiction-specific roles within their governance structures. At the same time, global workflows can be constructed by requiring coordinated participation across multiple jurisdictions. This allows organizations to enforce regional compliance requirements while maintaining unified operational processes.

CERTEN also supports delegation of jurisdictional authority to specialized providers. External compliance services can be integrated as participating authorities, contributing signatures that attest to regulatory validation without requiring their internal logic to be embedded within execution environments.

Data residency requirements are preserved by limiting the movement of sensitive data. Institutional systems perform evaluations locally, and only cryptographic attestations of participation are transmitted to the identity layer. This ensures that sensitive information remains within jurisdictional boundaries while still enabling global verification of governance outcomes.

Policy Vaults are designed to enforce jurisdictional requirements at execution by rejecting any transaction that does not include valid participation from required jurisdictional authorities. This is designed to make compliance enforced as a condition of execution rather than optional or advisory.

8.4 Legal and Interpretive Considerations

While CERTEN is engineered to align with regulatory frameworks such as UCC Article 12 and MiCA, the ultimate determination of compliance remains subject to judicial interpretation and evolving regulatory guidance.

The architecture is designed to provide the strongest possible technical foundation for demonstrating control, segregation, and auditability. However, institutions implementing CERTEN should obtain independent legal advice to ensure that their specific configurations and use cases meet applicable regulatory requirements.

9. Network Fees and Validator Participation

CERTEN charges a fee for each governance proof produced and verified on a destination chain. These fees are designed to fund the operation of the validator network that produces proofs and the on-chain anchoring that makes them verifiable. Fees are tied to operational events (each proof corresponds to a discrete, verifiable governance action) and are denominated in stable value units to provide predictable cost modeling for institutional users.

9.1 Fee Structure

The protocol is designed to support two service tiers, each corresponding to a different operational requirement.

On-cadence anchoring: approximately \$0.15 per proof. Governance proofs are aggregated over a configurable batching window (currently approximately 15 minutes on testnet). At the end of each window, the validator network constructs a Merkle tree of all proofs in the batch and anchors the Merkle root to the destination chain. The per-proof cost is amortized across all proofs in the batch. This mode is intended for routine operations such as scheduled settlements, periodic treasury movements, and batch processing workflows where minutes-scale latency is acceptable.

On-demand anchoring: approximately \$0.45 per proof. For time-sensitive operations, an immediate anchor request is designed to trigger a smaller dedicated batch submitted within approximately 30 seconds plus destination chain block time. This mode is intended for use cases such as collateral adjustments, margin calls, and time-critical financial operations where immediate finality matters more than cost amortization.

Both tiers produce the same on-chain proof format and are verified by the same destination-chain contracts. The mode used for a given operation is configured at request time. Fee amounts are denominated in stable value and are subject to change as the protocol matures.

9.2 Cost Independence from Governance Complexity

A key architectural property of the protocol is that proof verification cost is designed to be independent of governance complexity. Traditional multi-signature systems incur costs that scale with the number of participants and the complexity of authorization logic, particularly on congested networks. CERTEN performs governance resolution within the identity layer at near-zero marginal cost. Only the final proof verification occurs on the destination chain, where verification cost is designed to be fixed regardless of validator count or governance complexity.

A complex governance structure involving many participants, multiple roles, and layered approval requirements is designed to incur the same verification cost as a simple authorization. This is intended to support sophisticated institutional governance without scaling costs with governance complexity.

9.3 Validator Participation and Network Security Rewards

Validators are participants in the CERTEN network who run consensus, observe the Accumulate ledger, independently re-execute the proof pipeline, and produce BLS aggregate signatures attesting to verified proofs. The validator network is the backbone of the protocol's trust model: a Proof of Intent is only valid if at least 2/3 of validators (by voting power) attest to it.

Validator participation is designed to require staking CTEN, the protocol's native participation token. Staking serves as an economic bond that is intended to align validator behavior with correct protocol operation. A validator that fails to participate or that signs incorrect proofs is designed to have its stake reduced through protocol-defined accountability mechanisms.

Stakers are designed to receive service rewards for the network security service they provide. The protocol is designed to distribute a portion of proof fees as service rewards to stakers and validators. This reward structure is designed to align participant incentives with reliable network operation. Rewards are tied to actual network usage. The protocol is designed not to inflate the supply of CTEN to fund staker rewards.

CTEN may also be used within the protocol for fee payment. The protocol's design intent is to support stable-value-denominated pricing while enabling token-based participation for users who prefer it.

CTEN supply, distribution, and any vesting schedules are described in separate documentation and are not required to understand the operational architecture of the protocol.

10. Ecosystem and Use Cases

CERTEN enables a broad range of applications by providing a universal governance layer for digital assets. By separating governance from execution, the protocol allows institutions and applications to integrate complex, institutional-grade control without embedding policy logic into execution environments.

The system is designed to support both existing institutional workflows and new classes of applications that require coordinated control across multiple parties, systems, and jurisdictions.

10.1 Secured Lending and Collateral Management

Secured lending represents a primary use case for CERTEN.

In these scenarios, multiple parties must coordinate control over assets, including borrowers, lenders, custodians, and potentially third-party agents. Traditional approaches rely on custodial arrangements or contractual agreements to enforce control, introducing counterparty risk and limiting verifiability.

CERTEN enables deterministic enforcement of collateral control by requiring participation from all relevant authorities before assets can be moved. Governance structures can encode lending agreements, ensuring that collateral cannot be released, transferred, or rehypothecated without satisfying defined conditions.

This allows lenders to achieve verifiable control over collateral while allowing borrowers to retain operational participation within defined constraints.

10.2 Treasury Management

Institutional treasury operations require coordinated control across multiple stakeholders, including finance teams, risk management, and executive oversight.

CERTEN enables treasury workflows to be expressed directly within governance structures, allowing institutions to define approval hierarchies, thresholds, and operational controls without embedding logic into execution systems.

Transactions such as fund transfers, liquidity management, and investment allocation can be governed through structured participation, ensuring that all required approvals are obtained prior to execution.

10.3 Custody Orchestration

CERTEN enables orchestration across multiple custody systems without requiring assets to be consolidated into a single platform.

Institutions can maintain assets across different custodians, wallets, or blockchain environments while enforcing a unified governance model. Execution remains distributed, but control is centralized at the governance layer.

This allows institutions to avoid vendor lock-in while maintaining consistent control across all asset holdings.

10.4 Cross-Institution Workflows

Many financial operations require coordination between independent institutions, each with its own governance requirements.

CERTEN enables cross-institution workflows by allowing multiple governance structures to participate in the authorization of a single Intent. Each institution's authorities can contribute signatures according to their own policies, and execution occurs only when all required governance conditions across all parties have been satisfied.

This enables complex workflows such as syndicated lending, multi-party asset management, and coordinated settlement without requiring shared custody or centralized control.

10.5 Wallet-Integrated Governance

CERTEN integrates with existing wallet infrastructure by separating user interaction from governance resolution.

Wallets remain responsible for initiating transactions and managing user keys, while governance is handled by CERTEN. This allows applications to incorporate institutional-grade control without redesigning user interfaces or embedding complex logic into wallet software.

Users interact with familiar tools, while governance is enforced transparently through the protocol.

10.6 Emerging Application Classes

By providing a universal governance layer, CERTEN enables new categories of applications that were previously difficult or impossible to implement.

These include applications requiring coordinated control across multiple chains, compliance-gated execution, real-time governance in financial markets, and programmable institutional workflows that integrate human and system decision-making.

A growing class of these applications is agentic. In this design, autonomous agents manage treasuries, execute cross-chain strategies, and transact with other agents, each bounded by an external policy-engine signer within a scoped position in the governance hierarchy. An agent operator can run many agents, each controlling its own segregated account under a single governance hierarchy, with per-agent limits, expiry, and revocation. Because authority is resolved before execution and recorded as proof, an institution can delegate financial operations to agents while retaining provable, real-time control over what those agents are permitted to do. Section 5.5 describes the mechanism.

Because governance is resolved independently of execution, these applications can operate across heterogeneous environments without duplicating logic or sacrificing control fidelity.